
Technology transfer package — Toscanini invention

invfixturefluidics_hydraulic0e8d4aa7fcfb8 · run r541-ranked-pkg-battery-a · maturity
EARLY_TECHNICAL_EVALUATION · 3D class SYSTEM_3D

Problem definition

- Problem Id: fixture:fluidics_hydraulic:0
- Device: CSF shunt system
- Failure: proximal catheter obstruction by tissue
- Failure Mode: OBSTRUCTION
- Constraint: maintain long-term lumen patency and flow

Failure mode being addressed

6 failure modes recorded. Primary: OBSTRUCTION — trigger: the operating condition stated in the problem (mechanism detail beyond the problem statement and trigger quantification NOT ESTABLISHED)

Mechanism

Toscanini investigated the problem “inv:fixture:fluidics_hydraulic:0:e8d4aa7fcfb8” and, after 2 architecture generations of challenge-and-evolution, its current invention is generation 2. The causal mechanism is Porous microstructure resists fluid-path tissue ingrowth. The proposed intervention realizes this mechanism by placing porous titanium proximal catheter tip in the shunt lumen at the failure site identified in the problem statement, where the governing physical effect produces sustained fluid flow through the shunt valve under the stated constraint (maintain long-term lumen patency and flow); the transfer logic follows from the custodied source observation and its mechanism source sp...

Causal delta

What distinguishes this generation from its parent is the explicit causal change recorded in the evolution: Porous microstructure resists fluid-path tissue ingrowth. The proposed intervention realizes this mechanism by placing porous titanium proximal catheter tip in the shunt lumen at the failure site identified in the problem statement, where the governing physical effect produces sustained fluid flow through the shunt valve under the stated constraint (maintain long-term lumen patency and flow); the transfer logic follows from the custodied source observation and its mechanism source span. — the architecture delta versus the parent generation is recorded in the run state and inspectable i...

Architecture

- flow path / lumen architecture — status ENGINEERING_PROPOSED
- pressure regulation element — status ENGINEERING_PROPOSED
- sensing / feedback element (if closed-loop) — status ENGINEERING_PROPOSED
- termination interfaces (application-context-specific: source and sink boundaries) — status ENGINEERING_PROPOSED
- termination interfaces (patient / reservoir) — status ENGINEERING_PROPOSED

Subsystem breakdown

- flow path / lumen architecture — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)

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 - termination interfaces (application-context-specific: source and sink boundaries) — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)
 - termination interfaces (patient / reservoir) — status ENGINEERING_PROPOSED: NOT ESTABLISHED (requires design work)

Governing equations

- FLUID-001: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$ — Hagen-Poiseuille (laminar pipe flow)
 - FLUID-002: $Re = (\rho * v * D) / \mu$ — Reynolds number
 - FLUID-003: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$ — Orifice flow (sharp-edged restriction)
- (3 total; full bindings in EQUATION_REGISTRY.json)

Parameter definitions

- CP-001 lumen inner diameter [r m] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-002 lumen length [r m] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-003 surface roughness [NOT_ASSIGNED UNKNOWN] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-004 operating pressure head [ΔP Pa] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-005 flow rate range [Q m³/s] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-006 viscosity of flowed medium [μ Pa.s] = UNKNOWN (no sourced value) (UNKNOWN)
- CP-007 occlusion growth tolerance [NOT_ASSIGNED UNKNOWN] = UNKNOWN (no sourced value) (UNKNOWN)

Constraints

- Stated Constraint: maintain long-term lumen patency and flow
- Failure Mode Of Device: OBSTRUCTION

Engineering assumptions

- Assumption: mechanism transfers across domain context
- Basis: MODELLED
- Test: bench shunt flow loop; measure flow decay over 30 days
- Assumption: searched sources approximate the prior-art universe
- Basis: MODELLED
- Test: expand sources (see next_best_action)

Failure modes

- FM-009: OBSTRUCTION — severity UNKNOWN (no sourced severity basis)
- FM-DOM-001: proximal occlusion by tissue ingrowth — severity UNKNOWN (no sourced severity basis)
- FM-DOM-002: lumen collapse under external pressure (kink) — severity UNKNOWN (no sourced severity basis)
- FM-DOM-003: encrustation / mineral deposition — severity UNKNOWN (no sourced severity basis)
- FM-DOM-004: regime transition to turbulence altering calibration — severity UNKNOWN (no sourced severity basis)

severity basis)

- FM-DOM-005: check-valve seat wear opening reverse leakage — severity UNKNOWN (no sourced severity basis)

Verification plan

- VF-001: result NOT_TESTED — bench shunt flow loop; measure flow decay over 30 days
- VF-002: result NOT_TESTED — run killer experiment: falsification_test_from_candidate (EIG/cost 0.44573539382492045)
- VF-003: result NOT_TESTED — bench flow-loop characterization across operating range
- VF-004: result NOT_TESTED — pressure-drop mapping vs flow (identify regime transitions)
- VF-005: result NOT_TESTED — delivered-flow trend below baseline in a flow-loop or animal model; pressure-head rise at constant flow (measured quantity: forward patency / free flow area)
- VF-006: result NOT_TESTED — kink/collapse test across duty cycle; flow-stoppage signature under applied load (measured quantity: mechanical stress / fatigue endurance)
- VF-007: result NOT_TESTED — pressure-flow mapping across the full envelope showing regime-break curvature (measured quantity: pressure / pressure-drop)
- VF-008: result NOT_TESTED — reverse-leakage rate measurement after accelerated cycling (measured quantity: reverse-flow / leakage rate)

Build / manufacturing pathway

- WP-01: test article first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed); measurement physical bench/clinical experiment — NOT simulation
- WP-02: test article constraint-compliance envelope article; measurement constraint-compliance characterization
- WP-03: test article article for VF-001; measurement bench shunt flow loop; measure flow decay over 30 days
- WP-04: test article article for VF-002; measurement run killer experiment: falsification_test_from_candidate (EIG/cost 0.44573539382492045)
- WP-05: test article article for VF-003; measurement bench flow-loop characterization across operating range
- WP-06: test article article for VF-004; measurement pressure-drop mapping vs flow (identify regime transitions)
- WP-07: test article article for VF-005; measurement delivered-flow trend below baseline in a flow-loop or animal model; pressure-head rise at constant flow (measured quantity: forward patency / free flow area)
- WP-08: test article article for VF-006; measurement kink/collapse test across duty cycle; flow-stoppage signature under applied load (measured quantity: mechanical stress / fatigue endurance)
- WP-09: test article article for VF-007; measurement pressure-flow mapping across the full envelope showing regime-break curvature (measured quantity: pressure / pressure-drop)
- WP-10: test article article for VF-008; measurement reverse-leakage rate measurement after accelerated cycling (measured quantity: reverse-flow / leakage rate)

Regulatory posture

problem context (canonical applicability): MEDICAL_IN_VIVO. Context requirement [APPLICABLE]: medical-device regulatory pathway determination (FDA 510(k)/PMA or market jurisdiction equivalent) with biocompatibility evaluation (ISO 10993) and QMS (ISO 13485). regulatory pathway: UNKNOWN (no regulatory determination exists for this invention; per R370C correction the pathway is never inferred from device class alone)

4 candidate standards recorded (candidates, NOT compliance claims):

- ISO 13485 (medical QMS) (EXTERNAL_PRECEDENT_CANDIDATE; applicability verification is the buyer's/regulatory affair's first engineering action)
- ISO 14971 (medical risk management) (EXTERNAL_PRECEDENT_CANDIDATE; applicability verification is the buyer's/regulatory affair's first engineering action)
- ASTM F2394 (silicone elastomers for medical use) (EXTERNAL_PRECEDENT_CANDIDATE; applicability verification is the buyer's/regulatory affair's first engineering action)
- ASME B31 / EN 13445 (pressure piping and vessels, per installation jurisdiction) (EXTERNAL_PRECEDENT_CANDIDATE; applicability verification is the buyer's/regulatory affair's first engineering action)

no standard is claimed as met; every candidate carries verify-applicability before any submission decision

Decisive experiment

- Selected: falsification_test_from_candidate
- Eig: 0.44573539382492045
- Eig Per Cost: 0.44573539382492045
- Definition: physical bench/clinical experiment — NOT simulation
- Hypotheses: • H_effect_holds
- H_effect_fails
- Options Ranked: • falsification_test_from_candidate
- Epistemic Note: all priors MODEL_DERIVED; no EXPERIMENTALLY_ESTIMATED inputs exist at this stage (Art. XXVIII/XXXVIII: no synthetic shortcut to physical observation)

Expected measurements

- bench shunt flow loop; measure flow decay over 30 days
 - run killer experiment: falsification_test_from_candidate (EIG/cost 0.44573539382492045)
 - bench flow-loop characterization across operating range
 - pressure-drop mapping vs flow (identify regime transitions)
 - delivered-flow trend below baseline in a flow-loop or animal model; pressure-head rise at constant flow (measured quantity: forward patency / free flow area)
 - kink/collapse test across duty cycle; flow-stoppage signature under applied load (measured quantity: mechanical stress / fatigue endurance)
- (all NOT_TESTED — expected values are UNKNOWN until the experiment runs)

Acceptance criteria

Pre-registered rule (from VF-001): pre-registered decision rule fixed BEFORE the test: the measured effect must exceed the measured baseline under the identical setup by the pre-registration margin; the numeric margin is fixed from the measured baseline at pre-registration (no value is invented here — Art. VIII)

Unresolved technical risks

The run records its unknowns in the package's engineering definition; each unknown is material and none is suppressed. Physical output is NOT ESTABLISHED — no measurement of this invention exists.

3D explanation

A conceptual architecture visualization (SYSTEM_3D) — topology and named components only, NO engineering dimensions; engineering CAD is earned only when parameters are sourced (labeled in 3D_DESIGN_STATUS.json)

Artifact references

every layer of this package is named in PACKAGE_MANIFEST.json with per-file sha256; the machine-readable layers are ENGINEERING_TRACEABILITY, EQUATION_REGISTRY, UNKNOWN_ROADMAP, VALIDATION_ECONOMICS, MATURITY_BASIS, LOOP_STATE, PROVENANCE (top level) and the MODEL/ tree

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.